



Predicting daily milk yield for primiparous cows using data of within-herd relatives to capture genotype-by-environment interactions

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ABSTRACT

This study develops and illustrates a hybrid *k*-medoids, random forest, and support vector regression (K-R-S) approach for predicting the lactation curves of individual primiparous cows within a targeted environment using monthly milk production data from their dams and paternal siblings. The model simulation and evaluation were based on historical test-day (TD) milk production data from 2010 to 2016 for 260 Wisconsin dairy farms. Data from older paternal siblings and dams were used to create family units ($n = 6,400$) of individual calves, from which their future performance was predicted. Test-day milk yield (MY) records from 2010 to 2014 were used for model training, whereas monthly milk production records of Holstein calves born in 2014 were used for model evaluation. The K-R-S hybrid approach was used to generate MY predictions for 5 randomly selected batches of 320 primiparous cows, which were used to evaluate model performance at the individual cow level by cross-validation. Across all 5 batches, the mean absolute error and the root mean square error of the K-R-S predictions were lower (by 24.2 and 23.4%, respectively) than that of the mean daily MY of paternal siblings. The K-R-S predictions of TD MY were closer to actual values $74.2 \pm 2.0\%$ of the time, as compared with means of paternal siblings'. The correlation between actual TD MY and K-R-S predictions was greater (0.34 ± 0.01) than the correlation between the actual yield and the mean of paternal siblings (0.08 ± 0.01). The results of this study demonstrate the effectiveness of the K-R-S hybrid approach for predicting future first-lactation MY of dairy calves in management applications, such as milk production forecasting or decision-support simulation, using only monthly TD yields of within-herd relatives and in the absence of detailed genomic data.

Key words: milk production forecasting, dairy modeling, test-day model

INTRODUCTION

Lactation curves are graphical representations of the milk yield (MY) of individual cows or groups of cows according to DIM, and these are used widely for herd management and replacement decisions. Test-day (TD) models have been developed to describe the lactation curve (Wood, 1967; Wilmink, 1987; Olori et al., 1999; Mayeres et al., 2004), and in the case of primiparous cows, these models typically describe the general shape of the curve for fixed subsets of cows grouped by age at calving, season of calving, breed composition, or herd (Jamrozik and Schaeffer, 1997; Druet et al., 2003; Macciotta et al., 2004; Strabel and Jamrozik, 2006; Daltro et al., 2019). Recently, machine-learning algorithms have been used to forecast milk production with high accuracy, with the ability to adapt to different management conditions and capacity to provide predictions for individual cows (Murphy et al., 2014; Zhang et al., 2016; Nguyen et al., 2020). However, the lack of performance data from prior lactations makes prediction of TD yields and lactation curves for individual primiparous cows very challenging.

Despite the lack of historical data for a given new-born calf, TD records are typically available for siblings and other maternal and paternal relatives, and these data can contain important information about the local management and environmental conditions within a given herd (Meinert et al., 1988; Norman et al., 2005; Banos et al., 2007). In contrast to national genetic evaluation systems, which seek to remove the influence of local management and environmental conditions and provide estimated breeding values that are applicable for selection decisions in an average environment, tools that seek to inform management and replacement decisions at the individual herd level can benefit from knowledge of genotype-by-environment interactions specific to local conditions and herd management practices (Weigel et al., 2017). For example, Schultz and Weigel (2019) showed that including data of within-herd relatives can improve the accuracy of genomic predictions for feed efficiency in Holstein cattle. Studies from different countries have reported genotype-by-environment inter-

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actions for reproduction (Strandberg et al., 2009), feed efficiency (Yao et al., 2017), MY due to average herd protein yield and temperature-humidity index (Hayes et al., 2003), and overall climate (Tiezzi et al., 2017); whereas other studies reported that the genotype-by-environment effects between conventional and grazing farming systems could be negligible (Boettcher et al., 2003; van der Laak et al., 2016). Evidence supports the inclusion of genotype-by-environment interactions at the farm-level for farm-specific decision-support tools. Although some researchers have recently used DHI data to predict first-lactation performance, predictions were limited to a single TD and required as many as 39 predictor variables (Dallago et al., 2019), which might not be practical for routine herd management decisions. From a herd management perspective, predicted lactation curves and expected daily MY for individual animals that reflect herd management practices and local environmental conditions would enhance farmers' decision-making and financial planning (Mohd Nor et al., 2013; Jensen et al., 2018; Siewert et al., 2019). Because primiparous cows are managed, grouped, and monitored according to expected milk production, and because phenotypic production data for these animals is not yet available, new methods that could predict their lactation curves with reasonable accuracy would aid in proper planning and decision-making. Hence, we hypothesize that predictions of lactation curves and daily MY of primiparous cows, for use by dairy farmers in daily herd management decisions, can be improved by incorporating performance data of maternal and paternal relatives whose phenotypes have been measured in the same herd.

Therefore, the aim of this study was to develop and illustrate an architecture, comprised of machine-learning algorithms, for predicting the lactation curves of individual primiparous cows within a targeted environment using monthly milk production data from their dams and paternal siblings.

MATERIALS AND METHODS

Data and Family Unit

A filtered archive of 16,582 Holstein calves born in 2014 from all 260 dairy herds in Wisconsin, and their associated pedigrees, were obtained from a larger data set of TD records (format 4) from the Council on Dairy Cattle Breeding (Bowie, MD). The filtered data set included lactations of primiparous cows with a minimum first TD DIM of 5, a maximum TD DIM of 360, and at least 10 TD per lactation. Average MY for each TD are shown in Figure 1, along with the

number of observations for each TD used in model training and testing. Each lactation record contained a cow identification number, dam identification number, sire identification number, county code, herd number, date of birth, calving date, age at calving, lactation number, number of TD, MY, and DIM on TD. Records were organized according to the pedigree data. Sire information of each female calf was used to identify paternal siblings, which, along with the calf's dam, consisted of 6,400 family units of individual calves in the current study. Seventy-five percent of these 6,400 family units had 30 or fewer paternal siblings, 17% of them had 31 to 100 paternal siblings, and the rest 8% had between 101 and 500 paternal siblings. To balance data quantity and quality, during data preparation, paternal siblings were temporarily limited to the same geographic county and, therefore, the environmental variance was limited to herds located in the same Wisconsin county. In each family unit, the data of paternal siblings' MY records were collected from years 2010 to 2014, whereas the actual TD MY for female calves were collected from 2016.

Architecture of Analytical Models

We present a hybrid k -medoids, random forest (**RF**), and support vector regression (**SVR**) method, denoted as **K-R-S** approach, for predicting TD milk production records of primiparous cows. This hybrid structure was developed using an unsupervised learning k -medoids algorithm for clustering (Kaufman and Rousseeuw, 1990), supervised RF models for classification (Breiman, 2001), and SVR models for regression prediction (Smola and Schölkopf, 2004). The methodology and data workflow are illustrated in Figure 2.

Previous data mining studies have shown the importance of clustering, the task of grouping instances or data points into clusters such that data points within a cluster are similar to each other and dissimilar to those in other clusters (Velumugan and Santhanam, 2010). As shown in step 1 in Figure 2, records on a specific TD are fed into the k -medoids to identify clusters of similar MY. We tested different numbers of clusters and found that 10 clusters provided the best predictions. These clustered data are used to train RF and SVR models in subsequent steps; RF is a common and highly flexible machine-learning algorithm that has been used to predict MY at first TD (Dallago et al., 2019). In the present study, RF was used to classify future records into MY clusters and TD zones (step 2 in Figure 2). Each TD represents a continuous period of DIM, for which the boundary is determined by the minimum and maximum DIM from the clus-

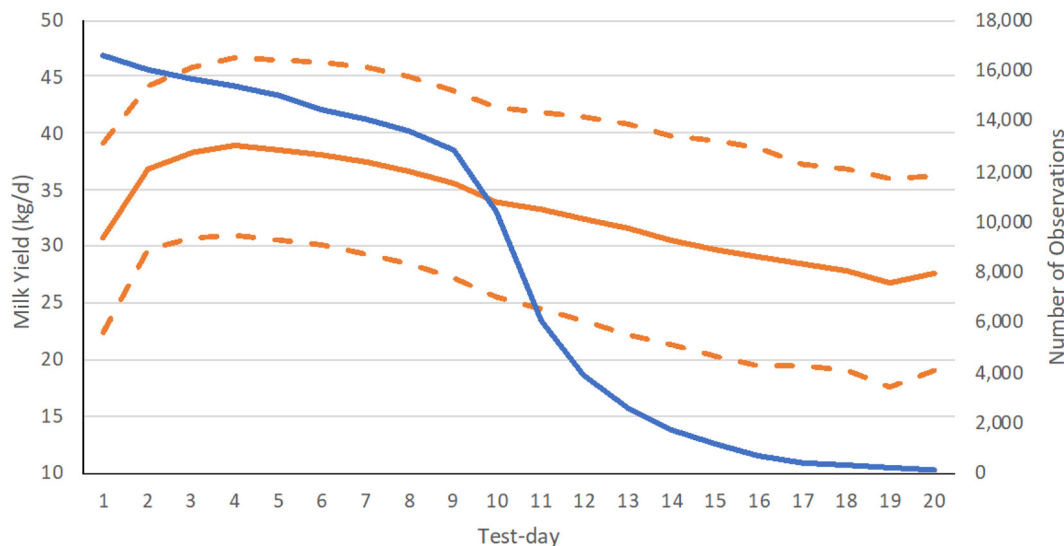


Figure 1. Characteristics of data set used in study. Means (amber continuous line) and SD (amber dashed lines) for test-day (TD) milk yields (kg/d, left y-axis) and number of observations (blue continuous line, right y-axis) recorded at different stages of lactation (20 TD, x-axis) for primiparous cows.

tered data (e.g., TD 4: DIM 94 to 124). The output of the RF model is a list probability of membership in different MY clusters [e.g., for cow ID 030304837273, the RF model gave a probability vector (0.101, 0.055, 0.314, 0.012, 0.078, 0.155, 0.062, 0.05, 0.102, 0.071) for clusters 1 to 10, respectively, on TD 4]. Each cluster and the probability vector were passed to the regression analysis in the next step, step 3. The SVR is a support vector machine algorithm for regression and function estimation that has been used to predict quantitative traits using, for example, genetic markers (Long et al., 2011) or daily MY records (Nguyen et al., 2020). As shown in step 3 in Figure 2, a separate local SVR model (regression layer) was developed for each MY cluster and TD to generate the milk production predictions. The SVR models can use the probability vector for different clusters generated by the RF model to produce a different MY prediction output for each cluster. We compared single predictions, which used the highest-ranking RF probability cluster, and weighted average predictions, which used all 10 clusters weighted by their probabilities. The output of the SVR models is the predicted MY for a given cow on one TD. The process was repeated 10 times to generate 10 discrete predictions for first-lactation TD MY for each calf. This simulation procedure was then repeated for all calves within a given herd.

The K-R-S hybrid approach described herein was implemented using the Scikit-learn (Pedregosa et al., 2011) library in Python 3.9 (Van Rossum and Drake, 2009) and MySQL database (Oracle Corporation, 2020).

Model Evaluation

Several statistical metrics were used to evaluate the performance of MY predictions: mean absolute error (**MAE**), root mean squared error (**RMSE**), coefficient of determination (**R²**), and correlation coefficient (**r**). The **R²** was used to evaluate the goodness-of-fit of nonlinear simulation for single animals, whereas the **r** was used to measure the strength and direction of linear relationships between predicted and actual TD MY observations at herd level. A nested cross-validation was selected to overcome the optimistically biased evaluation of the model performance, which contained 2 independent cross validations: an *i*-fold inner loop cross-validation during the model training and a *k*-fold outer loop cross-validation during the model evaluation, to nest the hyperparameter optimization procedure under the model selection procedure (Cawley and Talbot, 2010). Specifically, during steps 2 and 3 of the K-R-S (Figure 2), a 5-fold inner loop cross-validation was used to guarantee that the RF and SVR models were using the optimal estimator for each TD in each test batch. The aggregated and summary performance of predictions of the K-R-S hybrid approach was evaluated for 5 test batches of randomly selected animals, which was the 5-fold outer loop cross-validation. For each test batch, 320 family units (5% of entire family units) were randomly selected and compared against the predictions computed from the remainder of the data set.

The mean TD MY of older paternal siblings was used as a baseline prediction to compare with the single

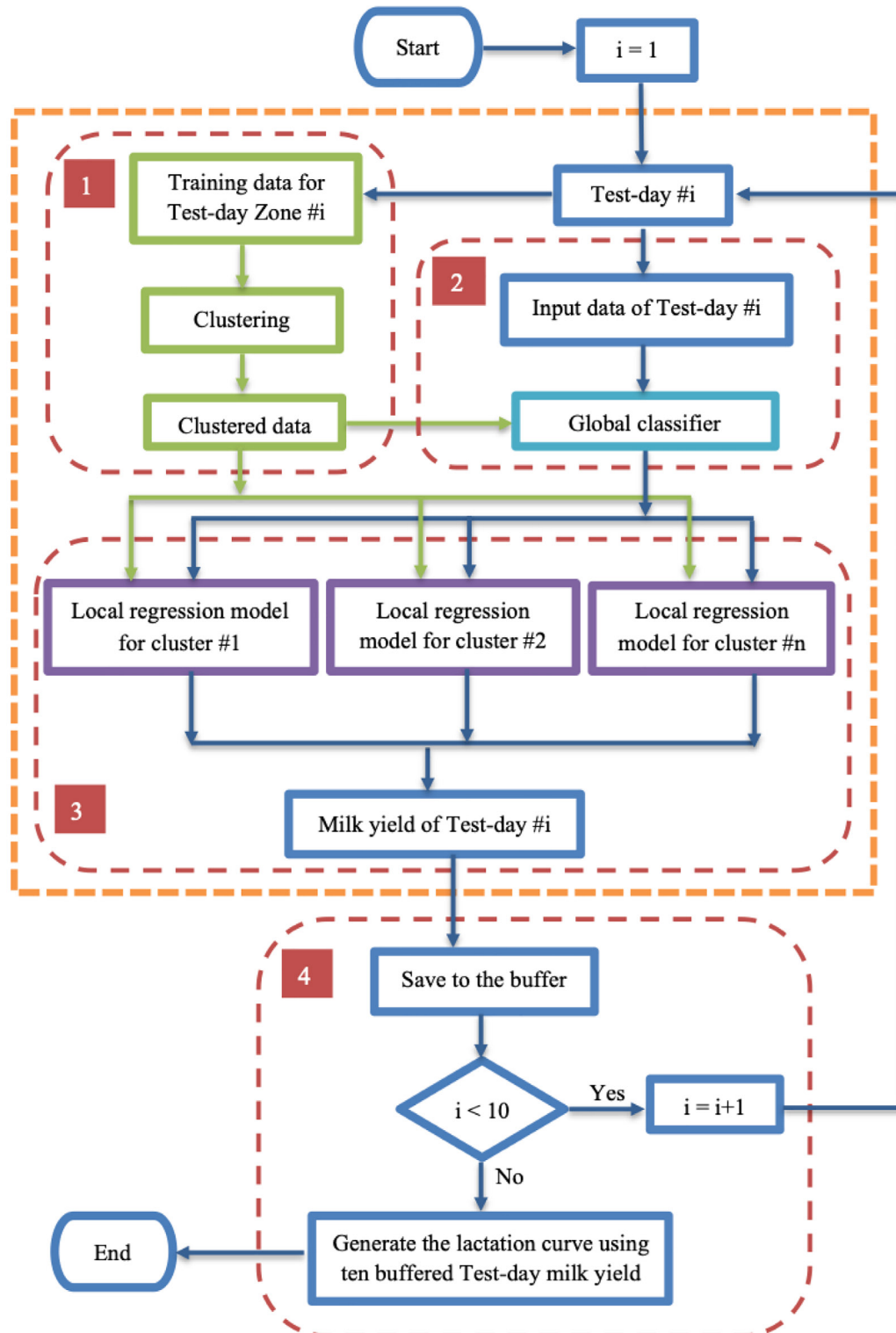


Figure 2. Analysis workflow for the proposed methodological *k*-medoids, random forest (RF), and support vector regression (SVR) hybrid (K-R-S) architecture. Test-day (TD) *i*, *i* is integer number of TD, range is 1 to 10; in local regression model, *n* is integer number of the cluster. It follows a 4-step workflow. Step 1 (upper left dashed line rectangle 1): perform a clustering model (*k*-medoids) using the historical data to identify clusters of similar milk yields (MY). Step 2 (upper right dashed line rectangle 2): train a RF model (classification layer) using the clustered data, which is used to classify the future MY into MY clusters and TD zones. Step 3 (middle dashed line rectangle 3): generate MY predictions using an independent local support vector machine model (regression layer) for each MY cluster on each TD zone. Step 4 (bottom dashed line rectangle 4): generate a lactation curve for 10 TD lactation outputs that consist of 10 discrete TD MY predicted by local SVR models.

Table 1. Aggregated actual and prediction performance of the *k*-medoids, random forest, and support vector regression (K-R-S) hybrid architecture (single and weighted average predictions) for first-lactation milk yield [MY; 10 test-days (TD)] for 5 batches of randomly selected animals (n = 320 per batch, 5% of records) and mean of paternal siblings for evaluation¹

Test batch	Average TD MY (kg/d)				MAE (kg)				RMSE (kg)				R ² ²			
	Total actual	Mean of paternal siblings			Mean of paternal siblings	Weighted average prediction			Mean of paternal siblings	Single prediction			Mean of paternal siblings	Single prediction		
		Mean	SD	Median		Mean	SD	Median		Mean	SD	Median		Mean	SD	Median
1	37.8	36.4	37.2	37.2	6.8	5.5	5.1	5.1	7.9	6.6	6.0	6.0	0.30	0.31	0.25	0.42
	6.1	4.1	4.1	2.7	3.4	2.5	2.6	2.6	3.5	2.7	2.8	2.8	0.25	0.25	0.28	0.27
	37.4	36.3	37.2	37.4	6.0	5.0	4.5	4.5	7.3	6.1	5.4	5.4	0.24	0.28	0.41	0.41
2	37.1	35.9	36.6	36.7	6.6	5.8	5.1	5.1	7.7	6.9	6.0	6.0	0.32	0.30	0.30	0.44
	5.8	4.0	4.3	2.9	3.0	2.3	2.4	2.4	3.2	2.6	2.6	2.6	0.27	0.25	0.28	0.28
	37.0	36.0	36.8	36.9	6.3	5.4	4.7	4.7	7.5	6.6	5.6	5.6	0.25	0.26	0.44	0.44
3	37.0	36.2	36.6	36.8	6.8	5.7	5.1	5.1	7.9	6.8	6.0	6.0	0.32	0.29	0.44	0.44
	5.9	4.1	4.4	3.0	3.2	2.5	2.5	2.5	3.3	2.8	2.7	2.7	0.27	0.24	0.28	0.28
	37.0	36.3	37.0	37.0	6.2	5.3	4.4	4.4	7.4	6.5	5.3	5.3	0.25	0.24	0.46	0.46
4	37.3	36.7	36.9	37.0	6.5	5.6	4.9	4.9	7.6	6.7	5.8	5.8	0.33	0.30	0.44	0.44
	5.6	4.1	4.5	2.9	3.1	2.4	2.3	2.3	3.4	2.6	2.5	2.5	0.26	0.26	0.29	0.29
	37.4	36.9	37.1	37.2	6.0	5.2	4.4	4.4	7.1	6.3	5.3	5.3	0.28	0.23	0.45	0.45
	37.3	36.7	37.2	37.2	6.4	5.6	4.9	4.9	7.6	6.7	5.8	5.8	0.31	0.30	0.42	0.42
	5.6	4.0	4.2	2.8	3.0	2.3	2.4	2.4	3.2	2.6	2.6	2.6	0.26	0.25	0.28	0.28
	37.3	36.9	37.5	37.3	5.9	5.3	4.4	4.4	7.1	6.3	5.3	5.3	0.25	0.26	0.43	0.43
Summary ³	37.3	36.4	36.9	37.0	6.6	5.7	5.0	5.0	7.7	6.8	5.9	5.9	0.31	0.30	0.43	0.43
	0.3	0.3	0.3	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.01	0.01	0.01	0.01

¹MAE = mean absolute error; RMSE = root mean squared error. Better predictions have lower MAE and RMSE, and higher R² value.
²The R² to evaluate the goodness-of-fit of nonlinear simulations ranges from 0 to 1; 1 indicates that the predictions perfectly fit the observed data (Sharma et al., 2007; Zhang et al., 2016; Migose et al., 2020).
³±SD of arithmetic mean.

Table 2. Summary performance of the *k*-medoids, random forest, and support vector regression (K-R-S) hybrid architecture (single and weighted average) for predicting entire first lactations for 5 batches of randomly selected animals (320 cows per batch, 5% of all records) compared with the mean of paternal siblings' milk yield

Test batch	Cows with better prediction using single prediction (%)	Cows with better prediction using weighted average prediction (%)
1	61.6	71.9
2	61.9	72.8
3	61.6	76.6
4	58.4	74.1
5	58.4	75.9
Summary ¹	60.4 (± 1.8)	74.2 (± 2.0)

¹ $\pm$ SD of arithmetic mean.

predictions and weighted average predictions from the hybrid K-R-S approach. The actual DIM of each TD MY was used as the DIM in generating TD MY predictions. Lactation curves were generated from predicted TD MY from the hybrid K-R-S approach using single predictions and weighted averages, and these were compared with actual TD MY data from each animal, as well as to the average TD MY of paternal siblings.

RESULTS

Results showed that our *k*-medoids, RF, and SVR hybrid architecture (K-R-S hybrid approach) consistently outperformed the mean of paternal siblings in predicting first-lactation TD MY of primiparous cows. Within the K-R-S approach, weighted average predictions were consistently closer to actual TD MY observations than single predictions derived from the most probably cluster (Tables 1 and 2).

Across all 5 test batches, the mean and median of MAE, RMSE, and R^2 of weighted average predictions from the K-R-S hybrid indicated superior predictive ability of this approach (Table 1). On average, the mean MAE of K-R-S approach single predictions was 13.6% (0.9 kg/d) lower than the mean MAE of the mean of paternal siblings' MY, and the mean of MAE of the K-R-S approach weighted average predictions was 24.2% (1.6 kg/d) lower than that of the mean of paternal siblings' MY. The mean of paternal siblings' MY had the highest RMSE values, followed by the K-R-S approach single predictions (11.7%, or 0.9 kg/d lower), and the K-R-S approach weighted average predictions (23.4% or 1.8 kg/d lower). The R^2 values of the K-R-S approach weighted average predictions were highest (closer to 1), followed by that of means of paternal siblings' MY and the K-R-S approach single predictions, clearly indicating closer proximity of the K-R-S weighted average predictions to the actual MY observations (Figure 3).

DISCUSSION

The purpose of this study was to develop and implement a hybrid machine-learning architecture for predicting lactation curves of individual primiparous cows, using readily available monthly TD milk production data from dams and paternal siblings, that had performed in the same herd under the same management and environmental conditions. This was achieved through a proposed machine-learning architecture described as the K-R-S hybrid approach. In contrast to predicting only first TD MY in a contemporary study (Dallago et al., 2019), the K-R-S hybrid approach provides the ability to construct a full lactation curve of primiparous cows for use in milk production forecasting, decision-support tools, and other herd management applications, even before the first data point is observed for a given animal.

Our machine-learning algorithms were trained and tested using monthly TD MY data that are widely available for all herds on DHI milk recording programs. However, the K-R-S hybrid structure developed herein, including the family unit filtering algorithm, can easily accommodate additional production data, such as daily TD MY records or weekly averages, or other key phenotypes, such as reproductive or health data. More frequent or more detailed data would allow a more detailed characterization of animals and their production environment, which would improve the prediction power of our proposed architecture.

Our K-R-S hybrid approach demonstrates a feasible solution for prediction of lactation curves in first-parity cows, for which performance data of prior lactations are unavailable. Interactions of genetic predisposition with local climate conditions and herd-specific management conditions can be incorporated by constructing family units on a within-herd basis. In situations where herds are too small to provide family units of adequate size, herds could be grouped geographically (e.g., by zip code or county), or they could be clustered into similar groups based on other demographic or performance data. Previous studies using traditional time-series models were unable to predict the performance of primiparous cows, because lactation data from the same cow in previous lactations is essential for model training (Murphy et al., 2014; Zhang et al., 2016; Nguyen et al., 2020). Likewise, predicting MY at first TD using a transition cow index model is also limited to multiparous cows and applicable only in the transition period (Schultz et al., 2016).

The K-R-S hybrid approach single prediction is the output from only one local SVR model in the regression layer, derived from the cluster with highest probability in the classification layer in the RF model. Hence, the

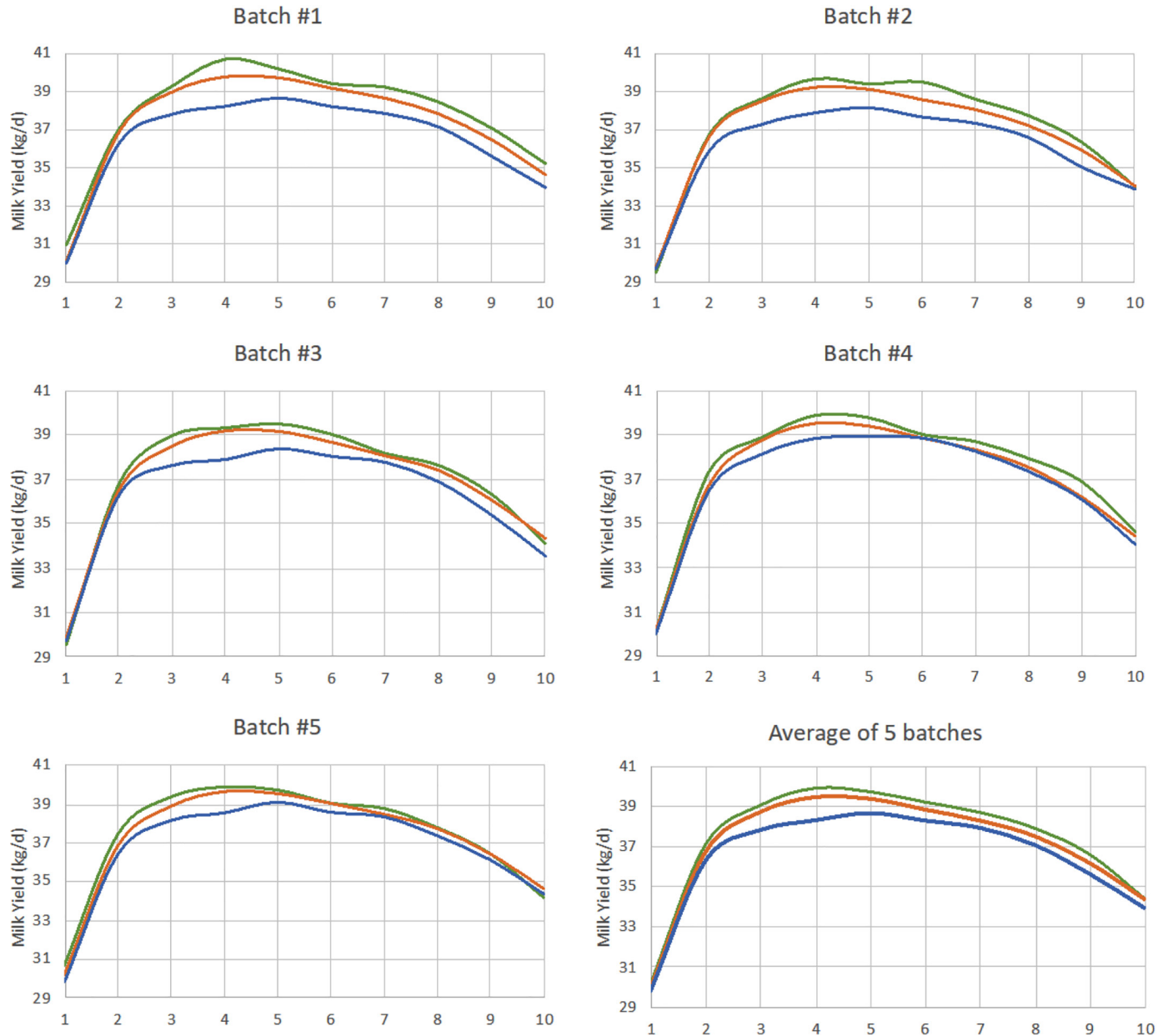


Figure 3. Recorded actual milk yield (MY, kg/d; green lines); mean of paternal siblings' MY (blue lines); and *k*-medoids, random forest, and support vector regression (K-R-S) hybrid architecture weighted average predictions (amber lines) at different stages of lactation (10 test-day; x-axis) for primiparous cows for 5 test batches of 320 cows per batch (5% of all records) and for the average of the 5 batches.

K-R-S single prediction is more variable in each round of calculations. Due to the flexibility of the K-R-S hybrid approach, a weighted average prediction can be generated without extra data requirements; the K-R-S weighted average prediction considered herein used the local SVR predictions in the regression layer and averaged them according to their probabilities. This reduced the noise associated with single-point predictions and provided more stable predictions with greater accuracy (Table 1). This feature is especially important

for first-lactation MY predictions, which are more variable than predictions for later lactations (Beauchemin and Rode, 1994; Jensen et al., 2018). The MAE and RMSE values in Table 1 are aggregated from individual cows, and their SD values represent the distribution around the mean values. Furthermore, the average performance of different predictions is shown in Table 2. These robust weighted average predictions resulted in lower SD, which indicates more precision (less variability) in predictions for the 320 cows in each test batch,

Table 3. The relationship (r) between actual test-day milk yield and the k -medoids, random forest, and support vector regression (K-R-S) hybrid architecture (single and weighted average) predictions and mean of paternal siblings over 5 test batches (320 cows per batch, 5% of all records)

Test batch	Mean of siblings	Single prediction	Weighted average prediction
1	0.08	0.27	0.33
2	0.09	0.25	0.36
3	0.07	0.24	0.33
4	0.07	0.25	0.34
5	0.08	0.24	0.33
Summary ¹	0.08 (± 0.01)	0.25 (± 0.01)	0.34 (± 0.01)

¹ $\pm$ SD of arithmetic mean.

coupled with greater prediction accuracy for individual cows. An independent comparison of the mean of paternal siblings' MY and the single or weighted average predictions presented in Table 3 confirms the consistently superior performance of K-R-S weighted average predictions, which could provide reliable outputs to be integrated into existing dairy farm decision-support systems.

The K-R-S approach weighted average predictions had an equivalent prediction accuracy to the mean of paternal siblings' MY over the entire lactation (Figure 3), despite the size reduction of the training data set for TD 9 and 10, when compared with TD 1 through 8 (Figure 1). The mean of paternal siblings' MY and the K-R-S hybrid approach predictions appear slightly lower than the actual TD MY for all 10 TD over 5 batches (Figure 3), indicating downward bias. This may be partially explained by positive genetic and management trends, because the data of paternal siblings' MY used in the predictions came from previous years (2010–2014), whereas the actual TD MY for new animals were from 2016. Genetic progress, together with improved management, would contribute to increases in average MY over time. Another possible reason is that the K-R-S hybrid approach treats every TD MY prediction independently. Hence, the size of training data set for one TD is not the same as for another. As depicted in Figure 1, the size of the training data set shows a gradually decreasing trend that reflects the actual situation of recorded lactation records, and this could reflect selection bias in the data. Primiparous cows could be removed from the herd any time after the first TD, due to low production, reproductive failure, health problems, or other reasons that might create bias in TD records from later in the lactation toward genetically superior animals that had not experienced any adverse health or management events. Such selection decisions would be made by individual farmers based on their own criteria, and these could be

inconsistent across time and between farms. Therefore, the K-R-S training data set contains, unavoidably, records from cows that were later removed for any issue, but only those cows that completed a minimum of 10 TD records were selected for testing purposes. Consequently, it is understandable that the K-R-S and the mean of paternal siblings had a small downward bias when compared with actual TD MY (Figure 3). This bias could have been partially removed by using only lactation data with a minimum 10 TD records for model training. However, we decided to include all data available from any TD to allow a larger training set and, more importantly, to better represent the on-farm realities of early culling and its effect on MY predictions.

Considering other recent studies that predicted only first TD MY for heifers (Schultz et al., 2016; Dallago et al., 2019), the K-R-S hybrid approach presents several advantages, including predicting MY for the first TD with relatively high accuracy (0.2 kg/d lower than actual MY), predictions for the entire first-lactation, and generating predictions for primiparous cows with no prior MY data. Other studies that predicted MY at the herd or the individual cow level required prior production records of the animals on which the prediction was made. Because primiparous cows do not have any prior MY records, those previous studies are not able to predict MY on primiparous cows (Murphy et al., 2014; Zhang et al., 2016; Nguyen et al., 2020). Those previous time-series or regression models required the output of a previous step as input of the current step on either training or testing to calibrate the trends. However, predictions should not use assumed existing historical data to train a model, because those data do not exist in reality. In contrast, the K-R-S hybrid approach is capable of predicting MY for primiparous cows, even when they do not have any prior MY records, which results in higher feasibility of practical use in herd management practices.

Albeit, the data available for the analysis (format 4, production data) was a limitation in this study, as more data streams are available either from the Council of Dairy Cattle Breeding or DHI processing centers (e.g., formats 5, 6, genetic evaluation, and so on), and we hypothesize that these will significantly improve our predictions. Additional breeding values and genomic test data would improve prediction accuracies, as it did when phenotypic information and genomic breeding values were combined to predict survival in Dutch dairy cattle (van der Heide et al., 2020). The K-R-S hybrid agile architecture allows us to easily merge other data sets beyond lactation data, such as health and reproduction data, as well as any other phenotypic traits of interest.

CONCLUSIONS

The proposed K-R-S hybrid approach in this study showed to be an efficient approach for milk production data processing, and prediction within a targeted environment, using available supplementary milk production data. The K-R-S hybrid approach proved to be an agile architecture that can perform simulation calculations for first-lactation MY prediction at the individual cow level, which automatically generates output results and analyses. The final deliverable is a computationally feasible tool which can perform frequent prediction simulations on farms and only require updated data of the individual farm. The K-R-S hybrid approach will be a useful benchmark platform for future model development and comparisons, because it is an integrated solution for data processing, model configuration, optimization, and analysis.

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